10/04/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

AIM – 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-2

Time : 180 Mins.

Answers

1. (1)	37. (4)	73. (3)	109. (4)	145. (2)
2. (2)	38. (1)	74. (3)	110. (1)	146. (4)
3. (3)	39. (2)	75. (2)	111. (3)	147. (4)
4. (2)	40. (1)	76. (3)	112. (4)	148. (4)
5. (2)	41. (1)	77. (4)	113. (4)	149. (3)
6. (1)	42. (4)	78. (3)	114. (2)	150. (4)
7. (2)	43. (2)	79. (4)	115. (3)	151. (3)
8. (1)	44. (3)	80. (3)	116. (2)	152. (1)
9. (4)	45. (4)	81. (3)	117. (4)	153. (3)
10. (4)	46. (2)	82. (1)	118. (4)	154. (3)
11. (2)	47. (2)	83. (3)	119. (3)	155. (2)
12. (1)	48. (3)	84. (1)	120. (4)	156. (2)
13. (2)	49. (3)	85. (1)	121. (4)	157. (2)
14. (3)	50. (2)	86. (1)	122. (3)	158. (4)
15. (2)	51. (2)	87. (4)	123. (3)	159. (3)
16. (2)	52. (3)	88. (1)	124. (3)	160. (4)
17. (3)	53. (4)	89. (3)	125. (2)	161. (2)
18. (2)	54. (1)	90. (3)	126. (2)	162. (4)
19. (3)	55. (1)	91. (4)	127. (4)	163. (2)
20. (2)	56. (2)	92. (4)	128. (2)	164. (4)
21. (3)	57. (2)	93. (2)	129. (2)	165. (4)
22. (4)	58. (3)	94. (1)	130. (2)	166. (3)
23. (2)	59. (2)	95. (4)	131. (2)	167. (2)
24. (1)	60. (1)	96. (2)	132. (4)	168. (3)
25. (2)	61. (2)	97. (2)	133. (2)	169. (2)
26. (4)	62. (1)	98. (2)	134. (3)	170. (2)
27. (3)	63. (3)	99. (3)	135. (4)	171. (2)
28. (4)	64. (1)	100. (2)	136. (4)	172. (4)
29. (2)	65. (2)	101. (3)	137. (3)	173. (3)
30. (1)	66. (4)	102. (3)	138. (2)	174. (3)
31. (4)	67. (4)	103. (4)	139. (2)	175. (4)
32. (2)	68. (3)	104. (1)	140. (2)	176. (4)
33. (1)	69. (3)	105. (2)	141. (1)	177. (3)
34. (2)	70. (1)	106. (2)	142. (2)	178. (2)
35. (4)	71. (2)	107. (2)	143. (4)	179. (3)
36. (1)	72. (4)	108. (3)	144. (4)	180. (4)

Hints and Solutions

CHEMISTRY

(1) **Answer :** (1)
Solution:
More stable is the alkene, lesser is the energy released on catalytic hydrogenation

(2) **Answer :** (2)
Solution:

$$\mathrm{CH_2 = CH_2} \xrightarrow[\substack{\mathrm{KMnO_4} \\ \mathrm{273\,K}}]{\text{dil.alkaline}} \underset{\text{Ethylene}}{\underset{\mathrm{OH}}{\mathrm{CH_2}}} - \underset{\text{glycol}}{\underset{\mathrm{OH}}{\mathrm{CH_2}}}$$

(3) **Answer :** (3)
Solution:

Nomenclature	IUPAC Official Name
Unniltrium	Lawrencium
Unnilhexium	Seaborgium
Unnilquadium	Rutherfordium
Unnilbium	Nobelium

(4) **Answer :** (2)
Solution:
For atomic number (Z) = 113,
Group = 13
Period = 7
If belongs to Boron family.

(5) **Answer :** (2)
Solution:
More will be the $E^\circ_{\text{Reduction}}$, lesser is the reducing power.

(6) **Answer :** (1)
Solution:
As electron withdrawing effect increases acidic strength of acid increases

(7) **Answer :** (2)
Solution:
$t_{1/2} = \frac{a_0}{2K}$
$\frac{a_0}{K} = 2t_{1/2}$
$t_{75\%} = \frac{3}{4}\frac{a_0}{K} = \frac{3}{4} \times 2t_{1/2}$
$= \frac{3}{2}t_{1/2}$
= 15 min

(8) **Answer :** (1)
Solution:
W = – P_{ext} ΔV
= – 1.5 (0.5)
= – 0.75 L bar
1L bar = 100 J
∴ –0.75 L bar = –75 J

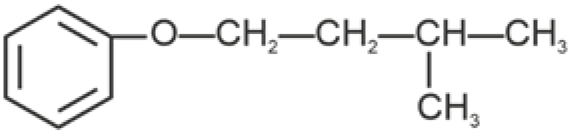
(9) **Answer :** (4)
Solution:
Mass of an atom of hydrogen = 1.6736 × 10^{-24} g
Nucleons in N_2 = 14 + 14 = 28

Nucleons in O_2 = 16 + 16 = 32

(10) **Answer :** (4)
Solution:
Conc. H_2SO_4 act as oxidising agent when it reacts with sulphur.
$S + 2H_2SO_4(\text{Conc.}) \rightarrow 3SO_2 + 2H_2O$

(11) **Answer :** (2)
Solution:
$\mathrm{Au} + 4\mathrm{H}^{+} + \mathrm{NO}_3^{-} + 4\,\mathrm{Cl}^{-} \rightarrow \mathrm{AuCl}_4^{-} + \mathrm{NO} + 2\mathrm{H_2O}$

(12) **Answer :** (1)
Solution:



3-Methylbutoxybenzene

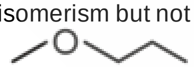
(13) **Answer :** (2)
Solution:
$\mathrm{Cr_2\,O_7^{2-}} + 3\,\mathrm{SO_3^{2-}} + 8\mathrm{H^+} \rightarrow 2\,\mathrm{Cr^{3+}} + 3\,\mathrm{SO_4^{2-}} + 4\mathrm{H_2O}$

(14) **Answer :** (3)
Solution:
• In Kjeldahl's method, nitrogen containing compound is heated with conc. H_2SO_4, then nitrogen present in it, gets converted into ammonium sulphate.
• Kjeldahl's method is not applicable to compounds containing nitrogen in nitro azo groups and nitrogen present in the ring as nitrogen in these compounds does not change to ammonium sulphate under these conditions

(15) **Answer :** (2)
Solution:

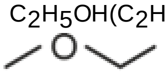
C_5H_{10}

⇒ This can show position isomerism but not metamerism

 &  ⇒ Metamers

Ethoxyethane Methoxypropane

$C_2H_5OH(C_2H_6O)$ does not show metamerism.



(C_3H_8O)

⇒ This can't show metamerisms

(16) **Answer :** (2)
Solution:
Six large chloride ions cannot be accommodated around Si^{4+} due to limitation of its size and hence interaction between lone pair of Cl^{-} ion and Si^{4+} ion is not very strong and $SiCl_6^{2-}$ does not exist.

(17) **Answer :** (3)
Solution:

Species	Hybridisation
XeO_3	sp^3
$[Ni(CN)_4]^{2-}$	dsp^2
SF_6	sp^3d^2
CO_3^{2-}	sp^2

(18) **Answer :** (2)

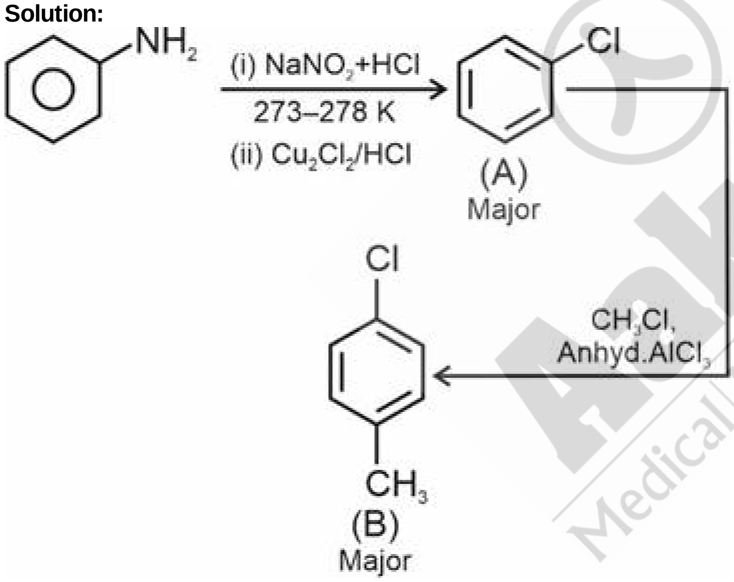
Solution:

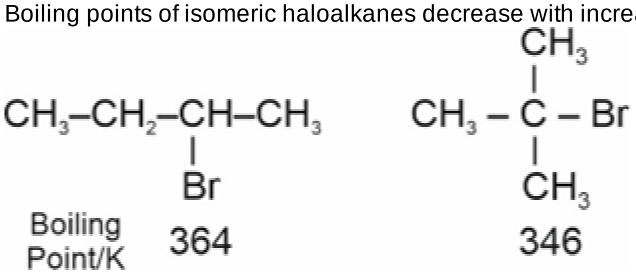
Compound	Bond angle
$BeCl_2$	180°
BF_3	120°
CH_4	109.5°
NH_3	107°
H_2O	104.5°

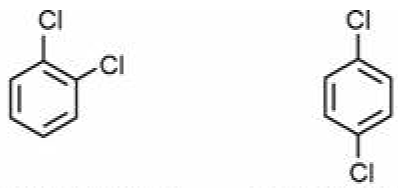
(19) Answer : (3)
Solution:
n-Hexane and n-pentane solution forms ideal solution.

(20) Answer : (2)
Solution:
$CaCl_2$ has maximum value of van't Hoff factor
So, ΔT_f is maximum
0.1 m $CaCl_2$ has minimum freezing point.

(21) Answer : (3)
Solution:
Complexes in which a metal is bound to only one kind of donar groups are known as homoleptic complexes
• $K_3[Co(C_2O_4)_3]$ is homoleptic while others are heteroleptic complexes

(22) Answer : (4)
Solution:


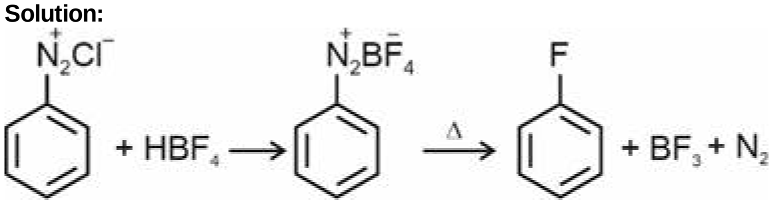
(23) Answer : (2)
Solution:
Boiling points of isomeric haloalkanes decrease with increase in branching




Boiling point/K	*o*-dichlorobenzene	*p*-dichlorobenzene
	453	488

For the same alkyl group, the boiling points of alkyl halides decrease in the order : RI > RBr > RCl > RF

(24) Answer : (1)
Solution:



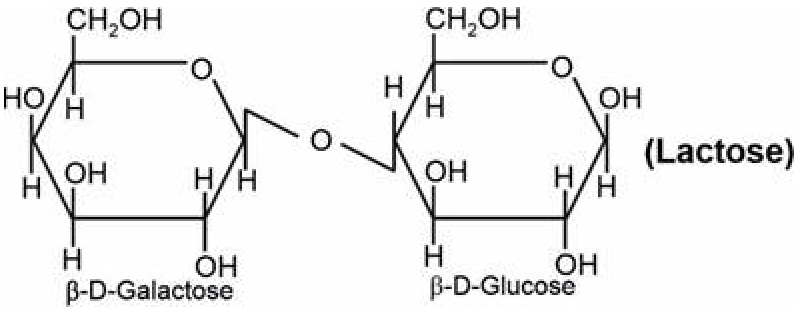
(25) Answer : (2)
Solution:
$Fe_4[Fe(CN)_6]_3$ – Prussian blue
PbS – Black
$(NH_4)_3PO_4 \cdot 12MoO_4$ – Yellow
$[Fe(SCN)]^{2+}$ – Blood red

(26) Answer : (4)
Solution:
$A + 2B \rightleftharpoons C, K_1$ …(i)
$E + 2F \rightleftharpoons B, \frac{1}{K_2}$
$2E + 4F \rightleftharpoons 2B, \left(\frac{1}{K_2}\right)^2$ …(ii)
Adding (i) & (ii),
$A + 2E + 4F \rightleftharpoons C, \quad K_1 \times \frac{1}{K_2^2}$
i.e., $\frac{K_1}{K_2^2}$

(27) Answer : (3)
Solution:
For acidic buffer,
$pH = pK_a + \log \frac{Salt}{Acid}$
$= 4.76 + \log\left(\frac{\frac{0.02\times50}{100}}{\frac{0.04\times50}{100}}\right)$
$= 4.76 + \log\frac{1}{2}$
= 4.76 – log 2
= 4.76 – 0.3010 = 4.459

(28) Answer : (4)
Solution:
$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{(Mg^{2+})}{(Ag^+)^2}$
$E_{cell} = 3.17 - \frac{0.0591}{2}\log\frac{(0.2)}{(0.1)^2}$
$E_{cell} = 3.17 - \frac{0.0591}{2}\left(1.3010\right)$
E_{cell} = 3.13 V

(29) Answer : (2)
Solution:



(30) **Answer :** (1)
Solution:

(31) **Answer :** (4)
Solution:
Radial node = (n-l-1)

Orbital	Radial node
4*p*	(4–1–1) = 2
5*s*	(5–0–1) = 4
4*d*	(4–2–1) = 1
4*f*	(4–3–1) = 0

(32) **Answer :** (2)
Solution:
Phase Transition are isothermal process.
$H_2O(l) \rightleftharpoons H_2O\ (g)$
$\Delta G = 0$ at equilibrium
$\Delta S_{system} > 0$
$\Delta S_{total} = 0$; at equilibrium

(33) **Answer :** (1)
Solution:
The equilibrium constant for an exothermic reaction decreases as the temperature increases.

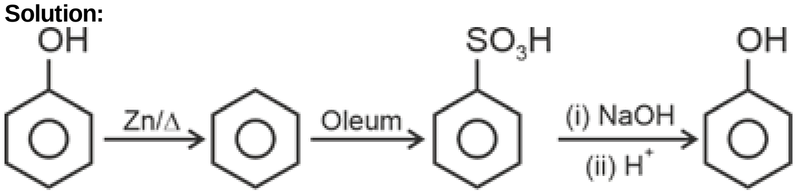
(34) **Answer :** (2)
Solution:

$NH_4NO_3 \xrightarrow{\text{Heat}} N_2O + 2H_2O$
$2NaNO_2 + 2FeSO_4 + 3H_2O \rightarrow Fe_2(SO_4)_3$
$+2NaHSO_4 + 2H_2O + 2NO$

$$2\,Pb\,(NO_3)_2 \xrightarrow{673\ K} 4\,NO_2 + 2PbO + O_2$$

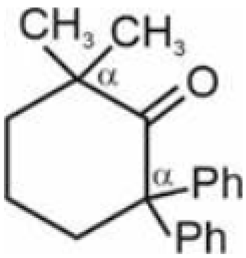
$$2NO + N_2O_4 \xrightarrow{250\ K} 2N_2O_3$$

(35) **Answer :** (4)
Solution:

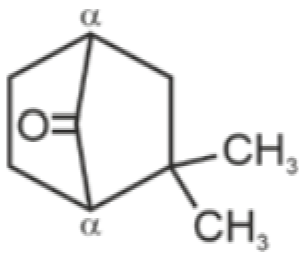


(36) **Answer :** (1)

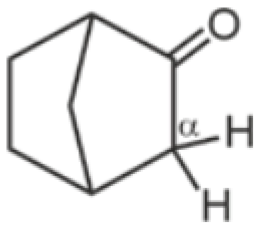
Solution:
Carbon atom at bridgehead does not contain enolisable hydrogen.



⇒ No enolisable hydrogen



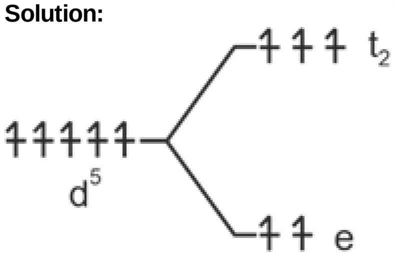
⇒ No enolisable hydrogen



⇒ 2 enolisable hydrogen

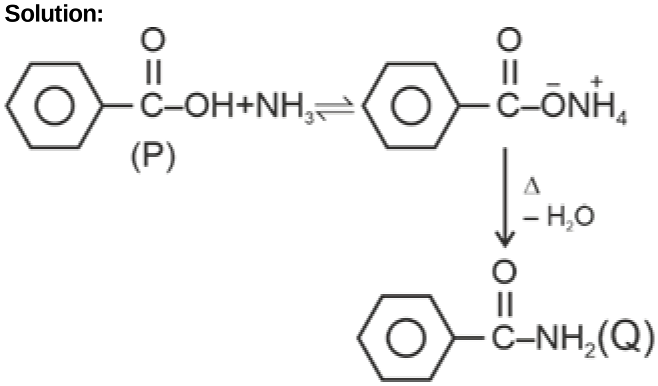
(37) **Answer :** (4)
Solution:
• O_2 Molecule has 2 unpaired electron So it is paramagnetic
• N_2^+ has bond order 2.5
• Li_2^{2+} has bond order zero so it doesnot exist

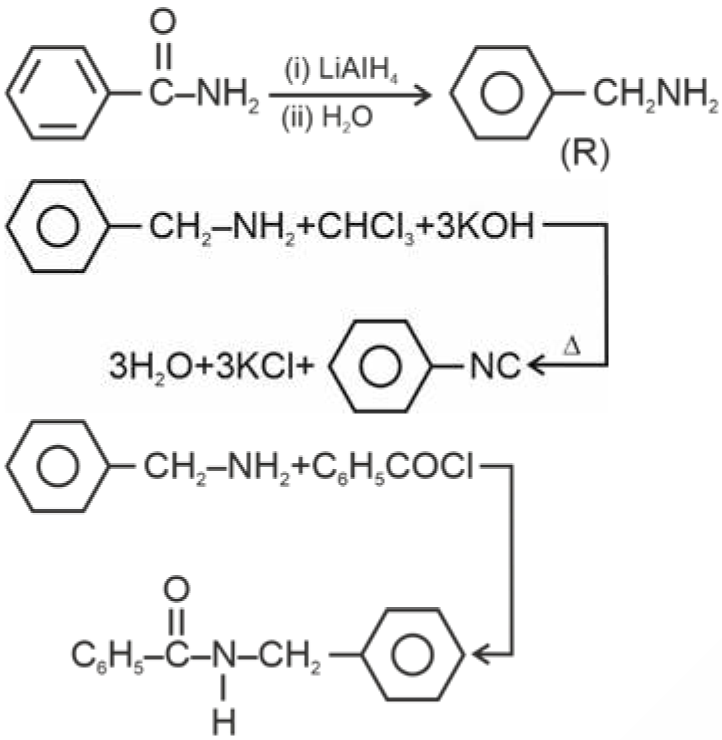
(38) **Answer :** (1)
Solution:



If has 5 unpaired electrons so its spin only magnetic moment is 5.9 BM

(39) **Answer :** (2)
Solution:





N-Benzylbenzamide

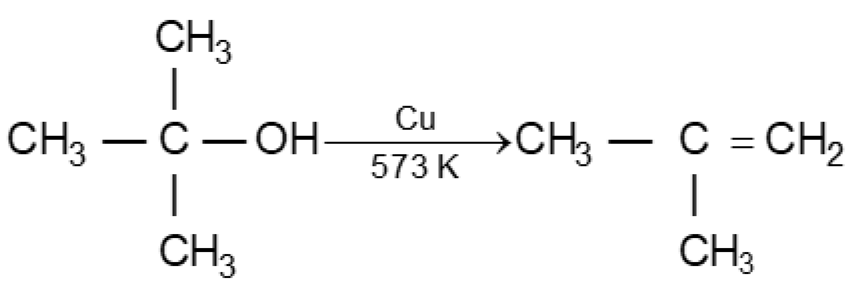
(40) **Answer :** (1)
Solution:
Threonine is an essential amino acid

(41) **Answer :** (1)
Solution:
n = 3, l = 1 means 3*p* subshell it can have maximum 6 electrons.

(42) **Answer :** (4)
Solution:
Ba^{2+} is a group V cation

(43) **Answer :** (2)
Solution:
$3\,MnO_4^{2-} + 4H^+ \rightarrow 2\,MnO_4^- + MnO_2 + 2H_2O$
V_2O_3 is Basic
V_2O_5 is amphoteric

(44) **Answer :** (3)
Solution:



(45) **Answer :** (4)
Solution:
The higher the n-factor of the compound, the more moles of $KMnO_4$ needed to react.
n-factor $FeSO_4$ = 1
FeC_2O_4 = 3

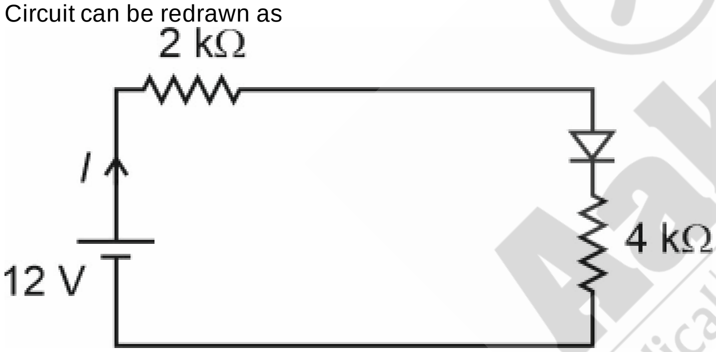
KI = 1
$Fe_2(C_2O_4)_3 = 6$

PHYSICS

(46) Answer : (2)
Solution:
$20VSD = 18MSD$
$1\,VSD = \frac{18}{20}MSD$
L.C. = $1MCD - 1VSD$
$= 1\,MSD - \frac{18}{20}MSD$
$= \frac{2}{20}MSD$
$= \frac{1}{10} \times 0.2$
= 0.02 mm

(47) Answer : (2)
Solution:
In single electron species like hydrogen atom,
$E \propto \frac{z^2}{n^2}$
since n is same for both
$\therefore \quad \frac{E_H}{E_{He}} = \frac{1}{2^2} = \frac{1}{4}$
$E_{He} = 4E_H = 4E$

(48) Answer : (3)
Solution:
Circuit can be redrawn as



$I = \frac{12}{6}$
= 2 mA

(49) Answer : (3)
Solution:
According to Einstein's equation
$(K.E)_{max} = hf - \varphi$

(50) Answer : (2)
Solution:
Band gap $= \frac{12400}{\lambda(\text{A})}$ eV
= 2 eV

(51) Answer : (2)
Solution:
As we know, $R \propto A^{1/3}$
$\frac{R_1}{R_2} = \left(\frac{8}{64}\right)^{1/3} = \left(\frac{1}{8}\right)^{1/3}$
$\frac{R_1}{R_2} = \frac{1}{2}$

$\Rightarrow \quad \frac{A_1}{A_2} = \left(\frac{R_1}{R_2}\right)^2 = \frac{1}{4}$

$\Rightarrow \quad \frac{A_0}{A} = \frac{1}{4} \quad \Rightarrow \quad A = 4A_0$

(52) Answer : (3)

Solution:

Solar cell works without any biasing.

• When pure silicon is doped with trivalent impurity, then it form p-type

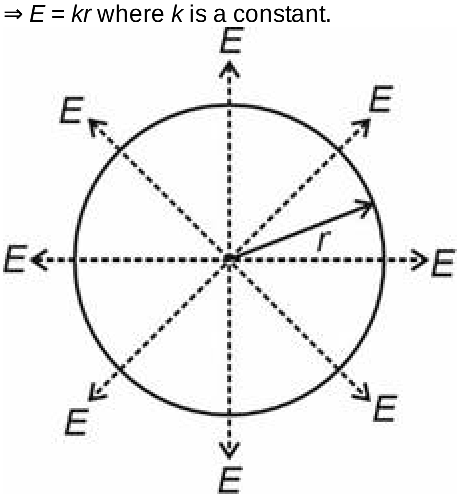
• In n-type semiconductor, $n_e >> n_h$

(53) Answer : (4)

Solution:

Given, $E \propto r$

$\Rightarrow E = kr$ where k is a constant.



Apply Gauss law:

$kr \times 4\pi r^2 = \frac{q_{in}}{\varepsilon_0}$

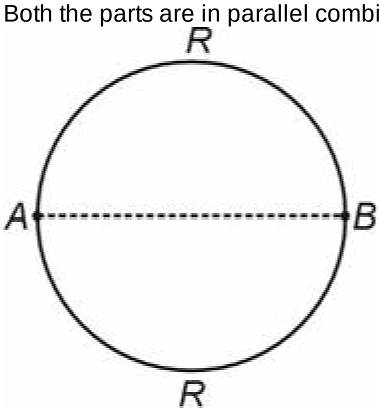
$\therefore \quad q_{in} \propto r^3$

(54) Answer : (1)

Solution:

Resistance of each half part will be $\frac{12}{2} = 6\ \Omega$.

Both the parts are in parallel combination,



hence $R_{eq} = \frac{6}{2} = 3\ \Omega$

(55) Answer : (1)

Solution:

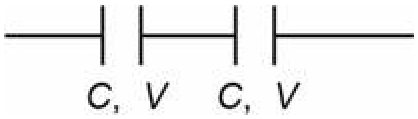
Emf is related to internal mechanism of a battery and it is non electrostatic in nature. Emf is equal to the work done by battery force per unit charge.

For a battery having emf E and internal resistance r, at the time of charging we can write terminal potential difference ΔV as

$\Delta V = E + ir \quad \Rightarrow \quad \Delta V > E$

(56) Answer : (2)

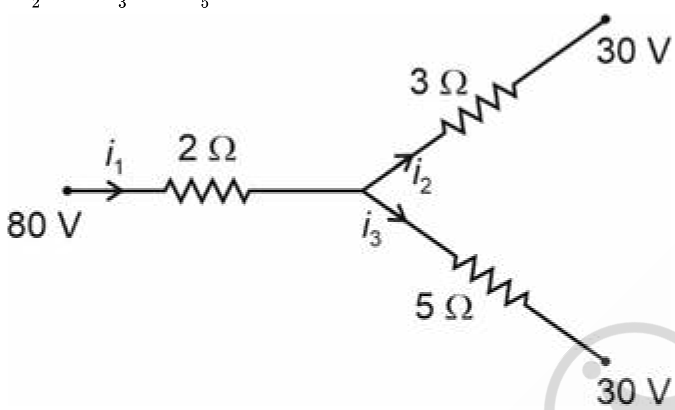
Solution:



In series, the equivalent capacitance will become $\frac{C}{2}$ and the breakdown voltage will get added, hence it will be $2V$

(57) Answer : (2)
Solution:
A neutral body can be charged positively if it loses electrons during rubbing, hence its weight should decrease slightly.

(58) Answer : (3)
Solution:
Apply Kirchhoff's junction rule:
$i_1 = i_2 + i_3$
$\frac{80-V_0}{2} = \frac{V_0-30}{3} + \frac{V_0-30}{5}$



$40 - \frac{V_0}{2} = \frac{V_0}{3} - 10 + \frac{V_0}{5} - 6$
$56 = \frac{V_0}{2} + \frac{V_0}{3} + \frac{V_0}{5} \Rightarrow 56 = \frac{15V_0+10V_0+6V_0}{30}$
$V_0 = \frac{56\times 30}{31} = \frac{1680}{31}$ Volt

(59) Answer : (2)
Solution:
Since $r >> a$, we can assume the system is made of two dipoles.
$P_1 = q \times 2a$, $P_2 = q \times 4a$
For both the dipoles, the point P is on axial point.
Potential for a dipole is given by
$V = \frac{KP\cos\theta}{r^2}$
$\therefore \quad V_T = V_1 + V_2 = \frac{KP_1\cos 0^\circ}{r^2} + \frac{KP_2\cos 0^\circ}{r^2}$
$V_T = \frac{6qa}{4\pi\varepsilon_0 r^2} = \frac{3qa}{2\pi\varepsilon_0 r^2}$

(60) Answer : (1)
Solution:
$\sin C = \frac{1}{\mu}$ and $\mu = a + \frac{b}{\lambda^2}$ ....
$\therefore \lambda\uparrow - \mu\downarrow$
Critical angle C is more for yellow as it has more wavelength.

(61) Answer : (2)
Solution:
Moment of inertia of rod about an axis perpendicular to rod passing through mid-point is $= \frac{ML^2}{12}$

(62) Answer : (1)
Solution:
Angle of deviation for a thin prism
$\Rightarrow \quad \delta = (\mu - 1)A$
$\Rightarrow \quad e - A = (\mu - 1)\,(A)$
$\Rightarrow A = e/\mu$

(63) **Answer :** (3)
Solution:
The law of conservation of angular momentum can be used to explain Kepler's 2nd law because the areal velocity of a planet $\left(\frac{dA}{dt}\right)$ is related to its angular momentum (L).
$\frac{dA}{dt} = \frac{L}{2m}$

(64) **Answer :** (1)
Solution:
For first lens, object is placed at 2*f* and hence, its image is formed at 20 cm behind this lens.
For second lens,
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-10} = \frac{1}{-10}$
$\Rightarrow \frac{1}{v} = \frac{-1}{10} - \frac{1}{10} = \frac{-2}{10} = \frac{-1}{5}$
$\therefore v = -5$ cm

(65) **Answer :** (2)
Solution:
At a depth *h* from the surface,
$g' = g\left(1 - \frac{h}{R}\right)$
Fractional change, $\frac{g-g'}{g} = \frac{-h}{R}$

(66) **Answer :** (4)
Solution:
The centre of mass (COM) depends on mass and positions of the particles as
$\vec{r}_{COM} = \frac{m_1\vec{r_1} + m_2\vec{r_2} + ...}{m_1 + m_2 + ...} = \frac{\sum m_i\vec{r_i}}{\sum m_i}$
The position of COM doesn't depend on reference frame but depends on the relative distance between the particles.

(67) **Answer :** (4)
Solution:
Power consumed in an A.C circuit containing only capacitor or only inductor will be zero.

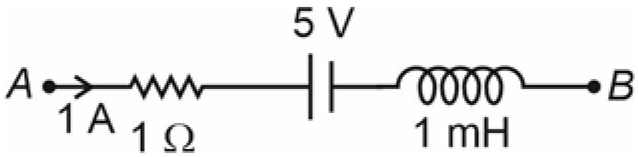
(68) **Answer :** (3)
Solution:
At $t = \infty$ inductor becomes short
So, $I = \frac{5}{20}$
$= \frac{1}{4}$ A

(69) **Answer :** (3)
Solution:
Velocity of EM wave in free space is given by, $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$

(70) **Answer :** (1)
Solution:
$B = \frac{\mu_0 Ni}{l}$
$B \propto \frac{N}{l}$
$\frac{B_A}{B_B} = \frac{N_A}{N_B} \times \frac{l_B}{l_A}$
$= \frac{N}{2N} \times \frac{3L}{2L}$
$= \frac{3}{4}$

(71) **Answer :** (2)
Solution:
$\tau = mlB\sin\theta$
$= mlB\sin 60°$
$= \frac{\sqrt{3}mBl}{2}$

(72) **Answer :** (4)
Solution:



$V_A - V_B = 1 + 5 - V_L$
4 = 6 – V_L
V_L = 2
$\frac{-Ldi}{dt} = 2$
$\frac{di}{dt} = (-2000 \text{ A/s})$

(73) Answer : (3)
Solution:
$\cos\phi = \frac{R}{Z}$
$\cos 60^\circ = \frac{R}{\sqrt{R^2+X_L^2}}$
$\sqrt{R^2+X_L^2} = 2R$
$R^2 + X_L^2 = 4R^2$
$X_L = \sqrt{3}R$

(74) Answer : (3)
Solution:
$\sqrt{PQ} = \sqrt{4} = 2$
$X = \sqrt{PQ}$
$\frac{\Delta X}{X} = \frac{1}{2}\left(\frac{\Delta P}{P} + \frac{\Delta Q}{Q}\right)$
$= \frac{1}{2}\left(\frac{0.4}{1} + \frac{0.2}{4}\right)$
ΔX = 0.45
= 0.45 m
$\therefore \sqrt{PQ} = (2 \pm 0.45)$ m

(75) Answer : (2)
Solution:
a = 5x, and $a = \frac{vdv}{dx}$
⇒ $vdv = 5xdx$
⇒ $\frac{v^2}{2} = \frac{5x^2}{2}$
⇒ $v \propto x$

(76) Answer : (3)
Solution:
V = 10L × 10L × 10L
= $10^3 L^3$
$V \propto L^3$
% change in volume $= \frac{\Delta V}{V} \times 100$
$= 3 \times \frac{\Delta L}{L}$
= 3 × 2%
= 6%

(77) Answer : (4)
Solution:
Fringe width, $\beta = \frac{D\lambda}{d},\ \beta \propto \lambda$
$\lambda_R > \lambda_G > \lambda_V$
$\therefore \beta_1 > \beta_2 > \beta_3$

(78) Answer : (3)
Solution:
In inelastic collision the kinetic energy conservation does not hold true.

(79) **Answer :** (4)
Solution:
Energy used = 100 W × 10
= 1000 W hr
= 1 Kilowatt hour
= 3.6×10^6 J

(80) **Answer :** (3)
Solution:
Distance in 4^{th} second
$S = u + \frac{a}{2}(2n - 1)$
$S = \frac{2}{2}(2 \times 4 - 1)$
S = 7 m

(81) **Answer :** (3)
Solution:
$V_{max} = A\omega = 5$
$a_{max} = A\omega^2 = 50\pi$
$\omega = 10\pi$
$T = \frac{2\pi}{\omega} = \frac{1}{5}$ s

(82) **Answer :** (1)
Solution:
$U = 5 \times \frac{5}{2}RT + 3 \times \frac{3}{2}RT$
$= \frac{25RT}{2} + \frac{9RT}{2} = \frac{34RT}{2} = 17RT$

(83) **Answer :** (3)
Solution:
Since $v \propto T \Rightarrow P$ is constants as $T\uparrow$ hence volume also increases at constant pressure.

(84) **Answer :** (1)
Solution:
The first law of thermodynamics is based on law of conservation of energy and temperature of body is not related to kinetic energy of motion of centre of mass.

(85) **Answer :** (1)
Solution:
Convection mode of heat transfer involves flow of matter within fluid due to temperature difference.
Change of solid directly into vapour is known as sublimation.

(86) **Answer :** (1)
Solution:
The pressure at the bottom must be same,
$\rho_A \times g \times 15 = \rho_B \times g \times 20 \quad \Rightarrow \quad \frac{\rho_B}{\rho_A} = \frac{15}{20} = \frac{3}{4}$

(87) **Answer :** (4)
Solution:
Speed of transverse wave on a stretched string = $\sqrt{\frac{T}{\mu}}$

Speed of longitudinal wave in a metallic bar = $\sqrt{\frac{Y}{\rho}}$

Speed of longitudinal wave in fluids, $V = \sqrt{\frac{B}{\rho}}$

(88) **Answer :** (1)
Solution:
From equation of continuity, we know that
$A_1V_1 = A_2V_2 + A_3V_3$
$\Rightarrow 16 = 12 + V_3 \Rightarrow V_3 = 4$ m/s

(89) **Answer :** (3)
Solution:
Maximum height attained, $H = \frac{u^2\sin^2\theta}{2g} = \frac{u^2}{4g}$

Horizontal range, $R=\frac{u^2\sin 2\theta}{g}=\frac{u^2}{g}$

∴ Required ratio $\frac{H}{R}=\frac{1}{4}$

(90) **Answer :** (3)
Solution:

Stress $=\frac{\text{Force}}{\text{Area}}=\frac{\text{N}}{\text{m}^2}$

Pressure $=\frac{\text{Force}}{\text{Area}}=\frac{\text{N}}{\text{m}^2}$

Stress is not a scalar quantity

BIOLOGY

(91) **Answer :** (4)
Solution:
Biological names are derived from Latin language irrespective of their origin.

(92) **Answer :** (4)
Solution:
Albugo candida, the parasitic fungi on mustard causes white spots on leaves. *Aspergillus* is known as weed of laboratory.

(93) **Answer :** (2)
Solution:
Sleeping sickness is caused by *Trypanosoma* which is a flagellated protozoan.

(94) **Answer :** (1)
Solution:

Opposite phyllotaxy	–	*Calotropis*
Alternate phyllotaxy	–	Mustard
Monoadelphous	–	China rose
Polyadelphous	–	Citrus

(95) **Answer :** (4)
Solution:
Dianthus has free central placentation.

(96) **Answer :** (2)
Solution:
Euglenoids have a protein rich layer called pellicle which makes their body flexible.

(97) **Answer :** (2)
Solution:
Datura has actinomorphic flower.

(98) **Answer :** (2)
Solution:
NADP reductase enzyme is located on the stroma side of thylakoid membrane.

(99) **Answer :** (3)
Solution:
Cyclic photophosphorylation occurs when light of wavelength beyond 680 nm is available for excitation. Cyclic photophosphorylation results only in the synthesis of ATP.

(100) **Answer :** (2)
Solution:
Earthworms are often considered as farmer's friend because they help in the breakdown of complex organic matter as well as loosening of soil.
Temperature and soil moisture are the most important climatic factors affecting the process of decomposition.

(101) **Answer :** (3)
Solution:

Complex I	NADH dehydrogenase
Complex II	Succinate dehydrogenase
Complex III	Cytochrome bc_1
Complex IV	Cytochrome c oxidase

(102) Answer : (3)
Solution:
Net primary productivity is the available biomass for the consumption to heterotrophs.

(103) Answer : (4)
Solution:
The conversion of glucose to glucose-6-phosphate is the first irreversible reaction of glycolysis and it is catalysed by hexokinase.

(104) Answer : (1)
Solution:
The earliest system of classification used only gross superficial morphological characters.

(105) Answer : (2)
Solution:
Selaginella and *Salvinia* are the members of pteridophytes which show the event that is precursor to seed habit.

(106) Answer : (2)
Solution:
To perform a chemical analysis, we can take any living tissue (a vegetable or a piece of liver, *etc.*) and grind it in trichloroacetic acid (Cl_3CCOOH) using a mortar and a pestle.

(107) Answer : (2)
Solution:
Based on number of amino and carboxyl groups, there are acidic (*e.g.*, aspartic acid, glutamic acid), basic (lysine) and neutral (valine) amino acids. Similarly, there are aromatic amino acids (tyrosine, phenylalanine, tryptophan).

(108) Answer : (3)
Solution:
Starch forms helical secondary structures. Starch can hold I_2 molecules in the helical portion. The starch-I_2 is blue in colour.
Glycogen is present as a store house of energy in animal tissues.

(109) Answer : (4)
Solution:
Collagen is the most abundant protein in animal world and Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

(110) Answer : (1)
Solution:
Columnar epithelium is found in the lining of stomach and intestine and it helps in secretion and absorption.

(111) Answer : (3)
Solution:
The cerebellum is the largest part of the hindbrain, and is sometimes called the little brain. It is located posterior to the brain stem at the back of brain.
Cerebellum has very convoluted surface in order to provide the additional space for many more neurons.

(112) Answer : (4)
Solution:
The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid *Salmonella typhimurium.*
Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance.

(113) Answer : (4)
Solution:
The convention for naming the restriction endonucleases is that, the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, *e.g.,Hin*d III comes from *Haemophilus influenzae* Rd. Thus, the letter 'd' is derived from the name of the strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria.

(114) Answer : (2)
Solution:

Taq polymerase has been isolated from *Thermus aquaticus* which is a thermophilic bacterium that can survive temperatures upto 95°C. In extension step, the *Taq* polymerase adds nucleotides to the primer, synthesizing a new DNA strand complementary to the template sequences.
It is a DNA dependent DNA polymerase.

(115) Answer : (3)
Solution:
The *Lac*-Z gene codes for the β-galactosidase. The enzyme utilises its substrate to produce blue coloured product. In the given case, the non-transformants will form white coloured colonies whereas, transformants can be
- Recombinants (blue coloured colonies)
- Non-recombinants (white coloured colonies)

(116) Answer : (2)
Solution:
The large holes in 'Swiss cheese' are due to the production of a large amount of CO_2 by a bacterium named, *Propionibacterium sharmanii.*

(117) Answer : (4)
Solution:
A population has certain attributes that an individual does not. An individual may have birth and death but a population has birth rates and death rates.

(118) Answer : (4)
Solution:
Nile perch was introduced in Lake Victoria which eventually led to elimination of species of cichlid fishes.

(119) Answer : (3)
Solution:
Both brown algae and red aglae:
(i) Perform vegetative reproduction by fragmentation.
(ii) Are found primarily in marine habitats.
(iii) Have cell wall composed of cellulose and hydrocolloids (algin and carrageen respectively)

(120) Answer : (4)
Solution:
Whisky, brandy and rum are produced by distillation of the fermented broth and they have higher ethanol concentration. Wine and beer are produced without distillation.

(121) Answer : (4)
Solution:
The age pyramid is bell-shape with pre-reproductive individuals being only marginally more than the reproductive individuals. Population is said to be mature or stable as there is no increase or decrease in its size due to growth rate being almost zero.

(122) Answer : (3)
Solution:
Endemic species are confined to a particular region and not found anywhere else.

(123) Answer : (3)
Solution:
Plant growth is generally indeterminate due to the presence of meristematic cells, which have the capacity to divide throughout the plant life. This form of growth, wherein new cells are always being added to the plant body by meristematic activity is called as open form of growth.

(124) Answer : (3)
Solution:
Spraying gibberellins on sugarcane crop increases the length of stem, thus increasing the yield of sugarcane.

(125) Answer : (2)
Solution:
Outer layer of a pollen grain (exine) has prominent apertures called germ pores where sporopollenin is absent.

(126) Answer : (2)
Solution:
Multicarpellary, apocarpous gynoecium is found in *Michelia*.

(127) Answer : (4)
Solution:
In angiosperms, egg apparatus consists of two synergids and one egg cell.

(128) Answer : (2)
Solution:
Haemophilia is a sex-linked recessive disorder in which a single protein that is a part of the cascade of proteins, involved in blood clotting is affected.

(129) Answer : (2)
Solution:
β-Thalassemia is controlled by a single gene HBB on chromosome 11.

(130) Answer : (2)
Solution:
Drosophila melanogaster produces large number of offsprings in a single mating. They shows clear differentiation of sexes and can be easily grown on simple synthetic medium in a lab.

(131) Answer : (2)
Solution:
Branchiostoma is a cephalochordate.
Petromyzon, *Scoliodon* and *Trygon* are vertebrates. All the mentioned animals are triploblastic and possess notochord which is a persistent throughout the life. *Branchiostoma* is non-vertebrate chordate.

(132) Answer : (4)
Solution:
Kidneys are bean shaped structures situated between the levels of last thoracic and third lumbar vertebra. Each kidney measures an average weight of 120-170 gms. Towards the centre of the inner concave surface of kidney is a notch called hilum through which ureter, blood vessels and nerves enter.

(133) Answer : (2)
Solution:
The theory of special creation has three connotations. One, that all living organisms that we see today were created as such. Two, that the diversity was always the same since creation and will be same in future also. Three, the Earth is about 4000 years old.

(134) Answer : (3)
Solution:
The universe is very old-almost 20 billion years old. The Big Bang theory attempts to explain to us the origin of Universe.

(135) Answer : (4)
Solution:
Darwin stated that any population has built in variation in characteristics. According to Darwin, the fitness refers ultimately and only to reproductive fitness. Hearts or brains of vertebrates are homologous structures.

(136) Answer : (4)
Solution:
Crustaceans like Prawns have antennal glands or green glands for excretion. Protonephridia or flame cells are the excretory structures in platyhelminths, rotifers, some annelids and the cephalochordates. Nephridia are the tubular excretory structures of earthworms and other annelids. Malpighian tubules are the excretory structures of insects.

(137) Answer : (3)
Solution:
Seed ferns were the ancestors of cycads.

(138) Answer : (2)
Solution:
The endocrine pancreas consists of "Islets of Langerhans". There are about 1-2 million Islets of Langerhans in a normal human pancreas representing only 1 to 2 percent of the pancreatic tissues.

(139) Answer : (2)
Solution:
Both malaria and amoebic dysentery are caused by protozoans. Female *Anopheles* is the vector for malaria and housefly acts as the mechanical carrier for transmission of amoebic dysentery. Amoebic dysentery is not a vector borne disease. Chills and high fever are seen in case of malaria while excess mucus and blood clots in stools are seen in case of amoebic dysentery.

(140) Answer : (2)
Solution:
In humans, four parathyroid glands are present on back side of thyroid gland. The parathyroid gland secretes a peptide hormone called parathyroid hormone (PTH). PTH stimulates the process of bone resorption (dissolution/ demineralisation).

(141) Answer : (1)
Solution:

A monomeric antibody contains four intrachain disulphide bonds in its light chains.

(142) Answer : (2)
Solution:
The glandular tissue of each breast is divided into 15-20 mammary lobes.
Several mammary ducts join to form a wider mammary ampulla which is connected to lactiferous duct through which milk is sucked out.

(143) Answer : (4)
Solution:
The cyclical changes in the ovary and uterus during menstrual cycle are induced by changes in the levels of pituitary and ovarian hormones.

(144) Answer : (4)
Solution:
Each ovum contains only X chromosome. So, it does not play any role in sex determination. The presence of either X or Y chromosome in the sperm determines the sex of the embryo.

(145) Answer : (2)
Solution:
The body of cockroach is segmented and divisible into three distinct regions -head, thorax and abdomen.

(146) Answer : (4)
Solution:
During pregnancy, all the hormones mentioned increase several folds except gonadotropins because increased levels of oestrogen and progesterone inhibit GnRH from hypothalamus that leads to decrease in secretion of gonadotropins (LH and FSH from pituitary). Gonadotropins have no role in maintenance of pregnancy.

(147) Answer : (4)
Solution:
In frog, fertilization and development is external and indirect respectively. Development takes place through a larval stage called tadpole larva.

(148) Answer : (4)
Solution:
Nutrient medium must provide a carbon source such as sucrose and also inorganic salts, vitamins, amino acids and growth regulators like auxins, cytokinins.

(149) Answer : (3)
Solution:
Bt toxin proteins exist as inactive protoxins but once an insect ingest the inactive toxin, it is converted into an active form of toxin due to alkaline pH of the gut of insect which solubilise the crystals.

(150) Answer : (4)
Solution:
The first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves during ventricular systole whereas the second heart sound (dub) is associated with the closure of semilunar valves during ventricular diastole. These sounds are of clinical diagnostic significance.

(151) Answer : (3)
Solution:
Right bundle branch and left bundle branch give rise to minute fibres throughout the ventricular musculature of the respective sides and are called Purkinje fibres. These are specialised muscle fibres.

(152) Answer : (1)
Solution:
Red muscle fibres contain plenty of mitochondria which can utilise the large amount of oxygen stored in them for ATP production. In white fibres, amount of sarcoplasmic reticulum is high.
Each meromyosin has two important parts, a globular head with a short arm and a tail, the former being called the heavy meromyosin (HMM) and the latter, the light meromyosin (LMM).

(153) Answer : (3)
Solution:
Hirudinaria (Blood sucking leech) is monoecious in nature and performs sexual reproduction.

(154) Answer : (3)
Solution:
The main challenge for production of insulin using rDNA technique was getting insulin assembled into a mature form. Chains A and B are produced separately in *E. coli* and disulphide bonds are produced between these chains.

(155) Answer : (2)
Solution:
Duration of each cardiac cycle = $\frac{60}{100} = 0.6$ second.

The atrium and ventricle of the same side are separated by a thick fibrous tissue called atrio-ventricular septum. A thin, muscular wall called the inter-atrial septum separates the right and left atria.

(156) Answer : (2)
Solution:
Morgan conducted dihybrid cross using *Drosophila* and found out that the proportion of parental gene combination was much higher than non parental gene combination.

(157) Answer : (2)
Solution:
Conjoint and open type of vascular bundles are found in dicot stem.

(158) Answer : (4)
Solution:
Bulliform cells are the modifications of adaxial epidermal cells in grasses. In dicot root, pith is small and inconspicuous. Chlorenchyma is chloroplast containing parenchyma.

(159) Answer : (3)
Solution:
Tobacco mosaic virus and QB bacteriophage have RNA as the genetic material.

(160) Answer : (4)
Solution:

Nirenberg and Matthaei	–	Used a synthetic poly U RNA and deciphered the code by translating this as polyphenylalanine
Taylor et.al.	–	Proved chromosomes replicate semi-conservatively in eukaryotes
Frederick Griffith	–	Discovered transforming principle by using two different strains of *Streptococcus*
Erwin Chargaff	–	Performed base composition studies for dsDNA

(161) Answer : (2)
Solution:
The presence introns is reminiscent of antiquity and the process of splicing represents the dominance of RNA world.

(162) Answer : (4)
Solution:
In exarch type of arrangement of primary xylem, metaxylem lies towards the centre. Trichomes in plants are usually multicellular. Guard cells are dumb-bell shaped in grasses.

(163) Answer : (2)
Solution:
Promoter gene provides the attachment site for RNA polymerase.

(164) Answer : (4)
Solution:
Genetic code is nearly universal, non-overlapping, commaless and degenerate.

(165) Answer : (4)
Solution:
Smooth endoplasmic reticulum is involved in the synthesis of lipids and steroids. SER plays an important role in detoxification of drugs.

(166) Answer : (3)
Solution:
Both mitochondria and chloroplast are double membrane bound structures and have dsDNA.
They are not the components of endomembrane system.

(167) Answer : (2)
Solution:
Diakinesis is the final sub-stage of prophase I, and it represents transition to metaphase I.

(168) Answer : (3)
Solution:
Congression of chromosomes takes place during metaphase stage.

(169) Answer : (2)

Solution:

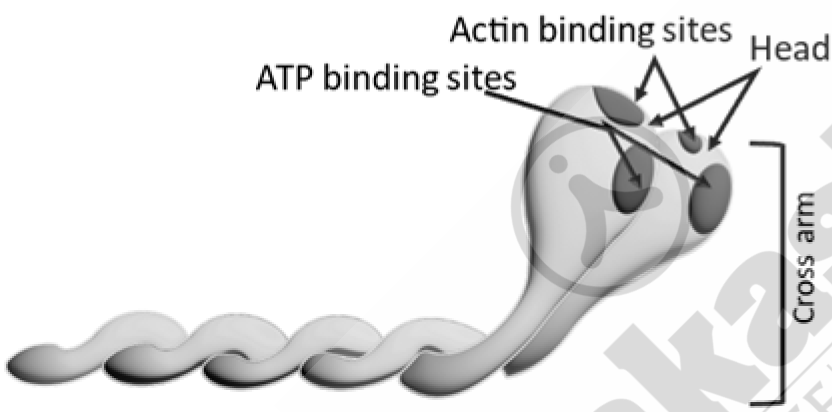
Synthesis of tubulin protein takes place	G_2 phase
Quiescent stage	G_0 phase
Interval between mitosis and initiation of DNA replication	G_1 phase
Duplication of centriole in cytoplasm	S phase

(170) Answer : (2)
Solution:
Lysosomes are rich in hydrolytic enzymes.

(171) Answer : (2)
Solution:
Tapeworms absorb nutrients from the host directly through their body surface as they do not possess alimentary canal.

(172) Answer : (4)
Solution:
Thoracic vertebrae, ribs and sternum together form the rib cage. Cervical vertebrae are not a part of rib cage.

(173) Answer : (3)
Solution:
Globular head is an active ATPase enzyme and has binding sites for ATP and active sites for actin.



(174) Answer : (3)
Solution:
Chondrocytes are cartilage cells which are enclosed in small cavities within the matrix secreted by them. Areolar tissue is a type of loose connective tissue.

(175) Answer : (4)
Solution:
Hydrolases: Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds.
Lyases: Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds.

(176) Answer : (4)
Solution:
MTP is performed to get rid of unwanted pregnancies either due to casual unprotected intercourse or failure of the contraceptive used during coitus or rapes. MTPs are also essential in certain cases where continuation of the pregnancy could be harmful or even fatal either to the mother or to the foetus or both.
In India, female foeticide is strictly prohibited.

(177) Answer : (3)
Solution:
Surgical methods, also called sterilisation, are generally advised for the male/female partner as a terminal method to prevent any more pregnancies. Surgical intervention blocks gamete transport and thereby prevents conception.
In tubectomy, a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through vagina. After tubectomy, menstruation still occurs but ova can not be received by fallopian tube. Estrogen production and ovum formation take place as it is.

(178) Answer : (2)
Solution:

A rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR), an increase in number of people in reproducible age as well as improved medical facilities are probable reasons for increase in population growth rate in India.

(179) **Answer :** (3)
Solution:
The diffusion membrane is made up of three major layers namely, the thin squamous epithelium of alveoli, the endothelium of alveolar capillaries and the basement substance (composed of a thin basement membrane supporting the squamous epithelium and the basement membrane surrounding the single layer endothelial cells of capillaries) in between them.

(180) **Answer :** (4)
Solution:
Vital Capacity (VC): The maximum volume of air a person can breathe in after a forced expiration. This includes ERV, TV and IRV or the maximum volume of air a person can breathe out after a forced inspiration.
TLC includes RV, ERV, TV and IRV or vital capacity and residual volume.